Problem 13. TIER 3 — A PRODUCT EQUAL TO A CONSTANT

At constant temperature, a gas obeys Boyle's law $PV = C$, where C is a constant. At one instant the volume is 600 cm^3, the pressure is 150 kPa, and the pressure is rising at 20 kPa/min. How fast is the volume changing?

$PV = \cancel{nRT}\ C$ $\quad P\uparrow, V\downarrow$

(1) no picture

(2) $V = 600$, $P = 150$, $\frac{dP}{dt} = 20$, $\frac{dV}{dt} = ?$

(3) $PV = C$

(4) $\frac{d}{dt}PV = \frac{d}{dt}C \rightarrow P\frac{dV}{dt} + V\frac{dP}{dt} = 0$

$$\frac{dV}{dt} = -\frac{V}{P}\frac{dP}{dt}$$

(5) $$\frac{dV}{dt} = \frac{-600}{150}(20) = -4(20) = -80\ \frac{cm^3}{min}$$